

Seokha Moon

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About Me

I am a Ph.D. student at Korea University's Vision & AI Lab, advised by Prof. Jinkyu Kim, and I hold a Bachelor's degree in Computer Science from Yonsei University. My research focuses on developing **autonomous driving algorithms, with broad interests spanning the full pipeline from perception to planning**. I have conducted research on perception models that effectively leverage spatio-temporal information, including 3D object detection and occupancy prediction, as well as trajectory prediction research that employs future-aware teacher-student learning to capture interaction dynamics and leverages vision-driven textual guidance to reason about agent context and intent.

Currently, I am interested in end-to-end frameworks for autonomous robots and driving, with a particular focus on causality-aware models and Vision-Language-Action (VLA) models that integrate perception, reasoning, and decision-making.

Education

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|----------------|--|------------------------|
| MS/Ph.D | Korea University , Computer Science and Engineering | Sep. 2022 to Present |
| | <ul style="list-style-type: none"> • GPA: 4.36/4.5 • Advisor: Prof. Jinkyu Kim | |
| BS | Yonsei University , Computer Science | Mar. 2016 to Aug. 2022 |
| | <ul style="list-style-type: none"> • GPA: Overall 3.45/4.3 Major 3.97/4.5 • Coursework: Operating System, Computer Architecture, Algorithms, Data Structures, Computer Graphics, Object-Oriented Programming, Artificial Intelligence, Computational Theory. | |

Experience

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| NAVER LABS , Robot & AD Localization, Research Intern | Gyeonggi-do, South Korea
Sep. 2024 to Mar. 2025
6 months |
| <ul style="list-style-type: none"> • Developed a efficient yet effective 3D occupancy prediction model (DisOcc). • Achieved SOTA performance while maintaining real-time inference speed and reducing memory usage by more than half compared to previous multi-frame fusion approaches. | |
| 42dot. , Motion prediction engineer, Intern | Seoul, South Korea
Aug. 2022 to Dec. 2022
4 months |
| <ul style="list-style-type: none"> • Developed algorithms to infer correlations between agents moving at intersections. • Researched a teacher-student model incorporating a student model that efficiently considers interactions between agents by mimicking teacher networks and teacher models that jointly consider the future movements of other agents. | |
| Korea Univ. Vision & AI Lab Internship , Intern | Seoul, South Korea
Jan. 2022 to Aug. 2022
9 months |
| <ul style="list-style-type: none"> • Implemented state of the art Multi-view 3D Object Detection model (ORA3D) • Developed algorithms for the occupancy prediction challenge using the Waymo Open Dataset. | |
| Samsung Electronics , Face Detection Project, Intern | Gyeonggi-do, South Korea
Mar. 2021 to June 2021
4 months |
| <ul style="list-style-type: none"> • Implemented few shot face detection algorithm using attention mechanism. • Created face dataset and designed a pipeline to augment dataset. | |

Publications

- ALIGN: Advanced Query Initialization with LiDAR-Image Guidance for Occlusion-Robust 3D Object Detection** 2026
Janghyun Baek, Mincheol Chang, [Seokha Moon](#), Seung Joon Lee, Jinkyu Kim
We introduce an object-aware, LiDAR-image guided query initialization strategy that significantly improves detection of occluded and crowded objects without modifying detector architectures.
arXiv (Under Review)
- Image-Guided Semantic Pseudo-LiDAR Point Generation for 3D Object Detection** 2026
Minseung Lee, [Seokha Moon](#), Seung Joon Lee, Reza Mahjourian, Jinkyu Kim
We introduce ImagePG, an image-guided pseudo-LiDAR point generation method that improves 3D object detection for sparse and distant objects while reducing false positives.
Winter Conference on Applications of Computer Vision (WACV)
- SUPER-AD: Semantic Uncertainty-aware Planning for End-to-End Robust Autonomous Driving** 2025
Wonjeong Ryu, Seungjun Yu, [Seokha Moon](#), Hojun Choi, Junsung Park, Jinkyu Kim, Hyun-jung Shim
SUPER-AD introduces uncertainty-aware BEV planning with lane-following priors for safer and more robust camera-only end-to-end autonomous driving.
arXiv (Under Review)
- Stream and Query-guided Feature Aggregation for Efficient and Effective 3D Occupancy Prediction** 2025
[Seokha Moon](#), Janghyun Baek, Giseop Kim, Jinkyu Kim, Sunwook Choi
We propose DuOcc, an efficient 3D occupancy prediction model that maintains dense voxel representations via dual aggregation with stream-based accumulation and query-guided refinement.
Work done during an internship at NAVER LABS.
arXiv (Under Review)
- Who Should Have Been Focused: Transferring Attention-based Knowledge from Future Observations for Trajectory Prediction** 2024
[Seokha Moon](#), Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim
We propose an attention-based teacher-student method that distills future-aware interaction knowledge to improve multi-agent trajectory prediction, yielding safer and more human-like predictions.
International Conference on Pattern Recognition (ICPR)
- VisionTrap: Vision-augmented Trajectory Prediction Guided by Textual Descriptions** 2024
[Seokha Moon](#), Hyun Woo, Hongbeen Park, Haeji Jung, Reza Mahjourian, Hyung-gun Chi, Hyerin Lim, Sangpil Kim and Jinkyu Kim
We introduce a novel approach incorporating visual features from surround-view cameras and textual descriptions for trajectory prediction.
European Conference on Computer Vision (ECCV)
- Learning Temporal Cues by Predicting Objects Move for Multi-camera 3D Object Detection** 2024
[Seokha Moon](#), Hongbeen Park, Jaekoo Lee, and Jinkyu Kim
We propose a Detection After Prediction (DAP) method to explicitly learn the temporal cues.
IEEE International Conference on Robotics and Automation (ICRA)
- BEVMap: Map-Aware BEV Modeling for 3D Perception** 2024
Mincheol Chang, [Seokha Moon](#), Reza Mahjourian and Jinkyu Kim
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

Enhancing Trajectory Prediction Accuracy with Goal Location and Lane Information <u>Seokha Moon</u> and Jinkyu Kim We introduce a method to predict the goal location and interact with lane to navigate towards the predicted goal. Conference on The Institute of Electronics and Information Engineers (IEIE) Oral	2023
RUI: Reducing Uncertainty in behavior prediction with Future Information <u>Seokha Moon</u> , Sejeong Lee, Hyun Woo, Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Vision-Centric Autonomous Driving	2023
ORA3D: Overlap Region Aware Multi-view 3D Object Detection Wonseok Roh, Gysam Chang, <u>Seokha Moon</u> , Giljoo Nam, Chanyoung Kim, Younghyun Kim, Jinkyu Kim and Sangpil Kim British Machine Vision Conference (BMVC)	2022

Teaching

Mathmatics for Computer Science (COSE111)	Spring 2024
Algorithm (COSE214)	Fall 2023
Algorithm (COSE214)	Fall 2022

Honors And Awards

First Prize of Computer Science , Yonsei University	Aug. 2021
HUAWEI scholarship , Talent Development Foundation	Nov. 2020

Skills

Languages: C++, C, Python, JavaScript

FRAMEWORKS & TOOLS: PyTorch, OpenCV, TensorFlow, Git, Visual Studio, XCode